FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – III
PAPER –1
TEST DATE: 18-01-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

Section – A ***(01 – 04, 20 – 23, 39 – 42)*****:** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A ***(05 –10, 24 – 29, 43 – 48)*****:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B ***(11 – 13, 30 – 32, 49 – 51)*****:** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

Section – C ***(14 – 19, 33 – 38, 52 – 57)*****:** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

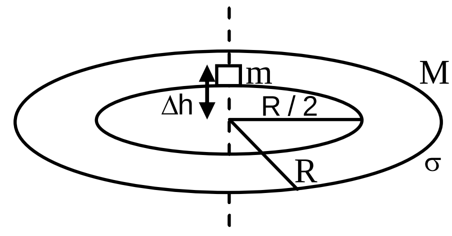
Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. A small block of mass m lies above a thin disk of total mass M, constant surface density σ, and radius R. A hole of radius R/2 is cut into the center of the disk, allowing the mass to pass though. The mass starts at a position along the central axis of the disk but is displaced a height Δh from the plane of the disk, where $\Delta h << R$. There are no other external forces at work on this system; i.e., the disk and block are out in deep space. Assume m << M. For the first order in Δh the block undergoes simple harmonic motion with respect to the centre of the plane of the disk, compute the period of oscillation of block.



(A) $T \simeq \sqrt{\frac{2\pi R}{G\sigma}}$

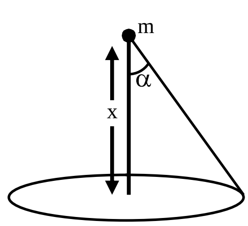
(B) $T \simeq \sqrt{\frac{\pi R}{G\sigma}}$

(C) $T = \sqrt{\frac{\pi R}{2G\sigma}}$

(D) None of these

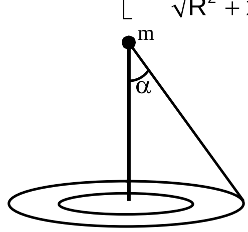
Ans. A

Sol.



$$g = 2\pi G\sigma\left[1-\cos\propto\right]$$

$$g = 2\pi G\sigma\left[1-\frac{x}{\sqrt{R^2+x^2}}\right] = 2\pi G\left[1-\frac{x}{R}\right]$$



$$g = 2\pi G\sigma\left[\left(1-\frac{x}{R}\right)-\left(1-\frac{x}{R/2}\right)\right]$$

$$g = \frac{2\pi G\sigma}{R}x$$

$$T = \sqrt{\frac{2\pi R}{G\sigma}}$$

2. Consider a metal can placed coaxially inside a long solenoid. The metal can be modelled as a thin cylindrical shell of radius r. The magnetic field due to the solenoid varies with time t as kt where k is a positive constant. The resistivity of the can is ρ and its breaking stress is σ. Ignore the self-inductance of the can and any currents induced in the flat surfaces of the can. Assume that the magnetic field produced by the induced currents is negligible. Compute the time at which the can breaks.

(A) $t = \frac{\sigma\rho}{r^2K^2}$

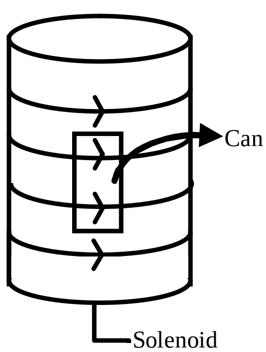
(B) $t = \frac{2\sigma\rho}{r^2K^2}$

(C) $t = \frac{\sigma\rho}{2r^2K^2}$

(D) $t = \frac{\sigma\rho}{3r^2K^2}$

Ans. B

Sol.

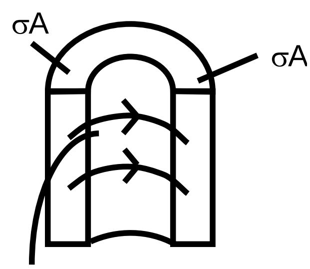


Emf is included in the can

$$Emf = \frac{dB}{dt}\pi r^2 = K\pi r^2$$

$$R = \frac{\rho \times 2\pi r}{A}$$

$$i = \frac{r}{2\rho}AK$$



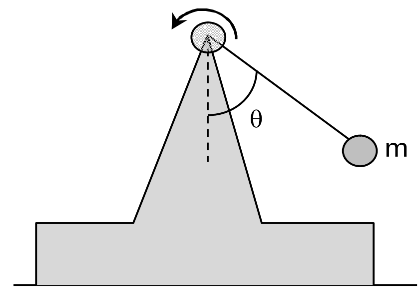
i2rB

From the diagram

$i \times 2rB = 2\sigma A$

$$t = \frac{2\sigma\rho}{r^2K^2}$$

3. A motor is installed at the top of a pole rigidly fixed on a platform. A light rod of length r = 1 m is rigidly attached to the motor shaft at its one end and at the other end a small ball of mass m is attached. The rod can rotated in a vertical plane with the help of the motor. Total mass of the platform, the pole and the motor is $\eta = 4$ times the mass of the ball. The motor rotates the rod at a constant angular velocity. The platform is placed on a horizontal surface, where coefficient of friction is $\mu = 1/\sqrt{3}.$ At which minimum angular velocity ω_0 of the rod, will the platform start sliding?



(A) $\omega_0 = 2\text{rad}/\text{s}$

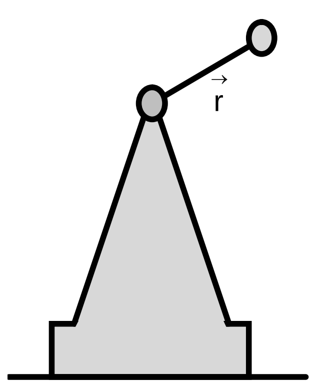
(B) $\omega_0 = 3\text{rad}/\text{s}$

(C) $\omega_0 = 5\text{rad}/\text{s}$

(D) None of these

Ans. C

Sol.

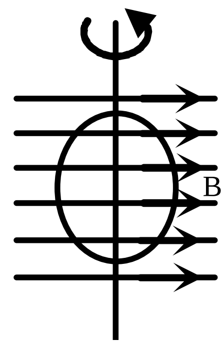


$\vec{F}$ on ball$+m\vec{g} = -mw^2\vec{r}$

$\vec{F}$ on platform$=m\vec{g}+mw^2\vec{r}$

$m\omega^2 r=\frac{\mu 5mg}{\sqrt{1+\mu^2}}$ $\omega_0=5 rad/s$

4. A ring of mean radius r and cross-sectional area A is made of a perfectly conducting wire. Inductance L of the ring is so small that inertia of free electrons cannot be neglected in the current building process. The free electron density in the conductor is n, mass of an electron is m and modulus of charge on an electron is e. Initially the ring is placed in a uniform magnetic field with its plane parallel to induction vector $\vec{B}$ of the field as shown in the figure. Find the current ring after it is turned through an angle 90°.



(A) $i=\frac{B\pi r^2}{L+\frac{m\times\pi r}{ne^2A}}$

(B) $i=\frac{B\pi r}{L+\frac{m\times\pi r}{ne^2A}}$

(C) $i=\frac{B\pi r^2}{L+\frac{m\times 2\pi r}{ne^2A}}$

(D) $i=\frac{B\times 2\pi r}{L+\frac{m\times\pi r^2}{ne^2A}}$

Ans. C

Sol. For perfect conductor R = 0 E_{net} = 0 in loop $\phi_{net}=$ constant

$\phi_{net}=\phi_{self}+\phi_k+\phi_{ext}$

$=L_i+L_{ki}+\vec{B}\cdot\vec{A}=L_{neti}+\vec{B}\cdot\vec{A}$

$L_K=\frac{ml}{ne^2A}$ (Where m = mass of one electron, l = loop length)

As ϕ is constant $\phi_i=\phi_f$

Initially $i=0$ & $\vec{B}\perp\vec{A}$

$\phi_{net}=0$

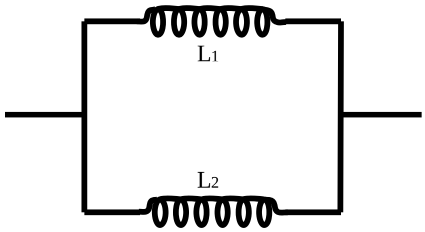
Finally $\phi=-BA'+L_{net}i$

$i=\frac{B\pi r^2}{L_{net}}=\frac{B\pi r^2}{L+\frac{m\times 2\pi r}{ne^2A}}$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

5.



For the given figure consider their mutual inductance is M then

(A) $Leq = \frac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$ When mutual inductance is not opposing

(B) $Leq = \frac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$ When mutual inductance is not opposing.

(C) $Leq = \frac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$ When mutual inductance is opposing.

(D) $Leq = \frac{L_1L_2 - M^2}{2L_1 + 3L_2 + 2M}$ When mutual inductance is opposing.

Ans. AC

Sol. $Leq = \frac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$ & $Leq = \frac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$ (Opposing)

6. In a photoelectric effect experiment, a point source of light of power 40W emits mono-energetic photons of wavelength λ_1 that can just exit photoelectrons from an isolated metallic sphere of radius 1 cm, placed at a distance of 1 km from the light source. Now, three other sources of wavelength λ_2, λ_3 and λ_4 which are emitting the same number of photons as that of λ_1 brought near the source of λ_1. Assume photo efficiency of 10^{-6} (Take $\lambda_1 = 4960\,\mathring{A}$, $\lambda_2 = 4133.33\,\mathring{A}$, $\lambda_3 = 5000\,\mathring{A}$, $\lambda_4 = 7200\,\mathring{A}$ and hc=12.400 eV$\cdot\mathring{A}$)

(A) Number of photoelectrons emitted from the sphere per second is 2.5×10^9

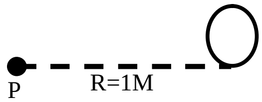
(B) The potential of the sphere when emission of photoelectron stops is 0.8 V.

(C) The potential of the sphere when emission of photoelectrons stops is 0.5 V.

(D) The time after which the emission of photoelectrons will stop is 1.39×10^{-3} sec.

Ans. ACD

Sol. $I = \frac{P}{4\pi R^2}$



Energy incident $= I\pi r^2$

No. of Photons incident/sec $= \frac{P \times \pi r^2}{4\pi R^2} \times \frac{\lambda}{hc}$

For λ_1 no. of photon incident/sec 2.5×10^{15}

Photo electric effect for λ_1 & λ_2 only.

$h\nu = \phi + K_{max} \quad \Rightarrow \quad K_{max} = 0 (\text{For } \lambda_1)$

$E_2 = \frac{12400}{4133.33} = 3ev$ $\phi = \frac{12400}{4960} = 2.5ev$

$V_s = 0.5 volt$

$\frac{KQ}{r} = V_s \Rightarrow Q_2 = ne$

7. A dog running with a constant speed v is chasing a cat that is running with a constant velocity $\vec{u}$. During the chase, the dog always heads towards the cat. At an instant, direction of motion of the dog makes angle θ with that of the cat and the distance between them is r.

(A) The magnitude of acceleration of dog at this instant is $\frac{uv\sin\theta}{r}$.

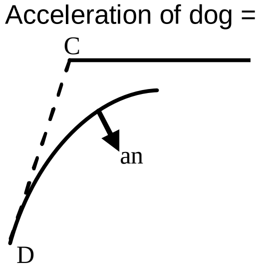
(B) Radius of curvature of path followed by dog at this instant is $\frac{vr}{u\sin\theta}$.

(C) The magnitude of acceleration of dog at this instant is $\frac{uv\cos\theta}{r}$.

(D) Radius of curvature of path followed by dog at this instant $\frac{ur}{v\cos\theta}$.

Ans. AB

Sol. Acceleration of dog = Normal acceleration.

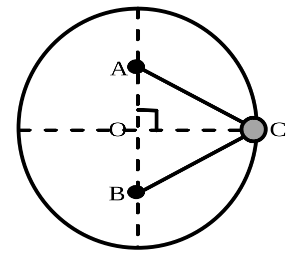


$\Delta A = \vec{A}_f - \vec{A}_i = \vec{A}_i + \Delta\vec{A}$.

$a = \frac{dv}{dt} = \frac{vd\theta}{dt} = v\omega = \frac{v\sin\theta}{r}$

$R = \frac{v^2}{a_n} = \frac{vr}{u\sin\theta}$.

8. A small bead of mass m can slide on a frictionless fixed ring of radius r. With the help of two identical strings of force constant k, the bead is connected to two nails A and B each on a diameter at a distance 0.5 r from centre O of the ring. Relaxed length of each string is negligible as compared to the radius of the ring. The bead is given a small velocity at point C. What can you predict about subsequent motion of the bead before any of the string strikes a nail? [ignore gravity]



(A) It will keep moving with its initial speed.

(B) It will oscillate simple harmonically about the point C.

(C) Its angular momentum about centre O of the ring is conserved.

(D) Total elastic potential energy stored in both the strings is $1.25kr^2$.

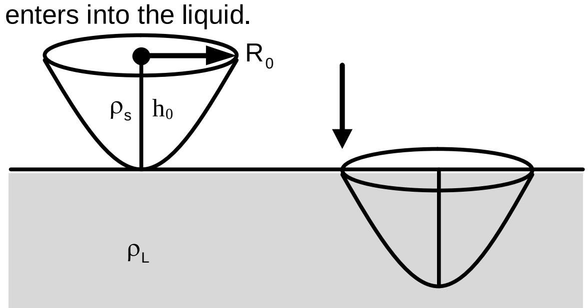
Ans. AC

Sol. $\vec{F} = -2\,k\,\vec{r}$ toward 0.

$\vec{\tau}_0 = 0$

$$\text{P.t} = \frac{1}{2}k\left(r^2 + \frac{r^2}{4}\right) = 1.25kr^2$$

9. A solid paraboloid of base radius R_0 and height h_0 is having uniform volume mass density of ρ_s is Inverted and placed just above the surface as shown, of a liquid having volume mass density of ρ_L. When paraboloid is released from rest it has been observed that when it becomes completely submerged in the liquid the paraboloid comes to the rest again. There exist uniform gravity of g. Assume that the liquid body is very large so that liquid level is not changing as the paraboloid enters into the liquid.



(A) The phenomenon is true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.

(B) The phenomenon is not true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.

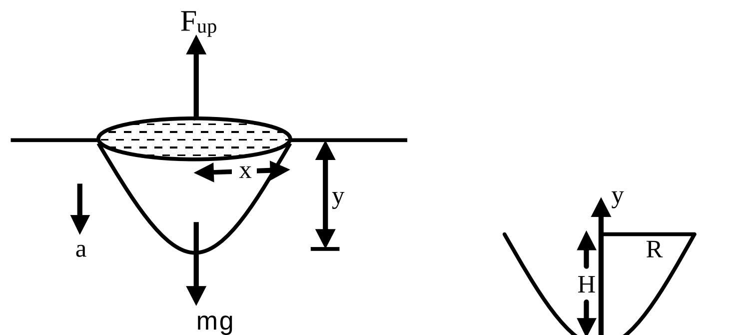
(C) The ratio of $\frac{\rho_L}{\rho_s}$ is 2.

(D) The ratio of $\frac{\rho_L}{\rho_s}$ is 3.

Ans. AD

Sol. Let $Y = ax^2$ (A parabola)

$Y = \frac{H}{R^2}x^2$ (from given data)



$$v = \frac{dy}{dt}$$

$$mg - \rho_L \cdot \frac{1}{2}\pi x^2 y \cdot g = m \cdot \frac{dv}{dt}$$

$$\frac{dv}{dt} = g - \frac{\pi x^2 y g \rho_L}{2\rho_s \frac{\pi R^2 H}{2}} g \Rightarrow g - \frac{\rho_L}{\rho_s}\frac{x^2}{R^2} \times \frac{yg}{H}$$

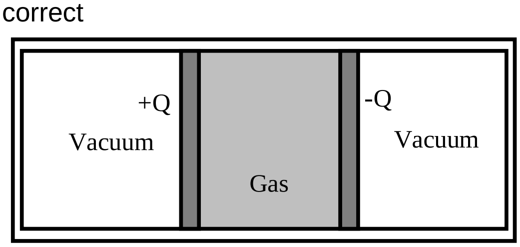
$$v \cdot \frac{dv}{dy} = g - \frac{\rho_L}{\rho_s} g y^2$$

$$\int_0^0 v dv = \int_0^H (g - \frac{\rho_L}{\rho_s} g y^2)\, dy$$

$$o = gH - \frac{g\rho_L}{3\rho_s}H$$

$$\frac{\rho_L}{\rho_s} = 3$$

10. A ideal gas having **f** degrees of freedom is kept inside a thermally insulated vessel, the vessel has two large pistons each with charge of +Q and -Q. Assume charges are uniformly distributed. Pistons can move without friction inside the vessel. At an instant the charge of pistons is increased **k** times with the help of an external agent. Then which of the following statements are correct



(A) The Ratio of final to initial temperature of gas in equilibrium is $\frac{K^2 + f}{2 + f}$.

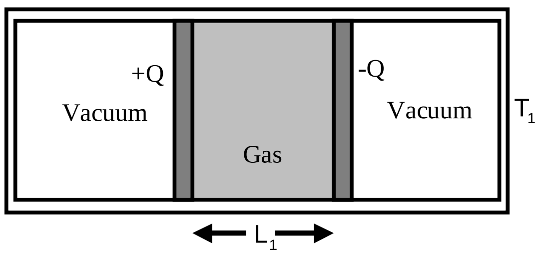
(B) The Ratio of final to initial temperature of gas in equilibrium is $\frac{2K^2 + f}{2 + f}$.

(C) Work done by the agent in the process is $(K^2-1)nRT_1$ where T_1 is initial temperature.

(D) Work done by the agent in the process is $\left(K^2-1\right)\frac{Q^2L_1}{A\varepsilon_0}$ where L_1 is initial separation between pistons.

Ans. BCD

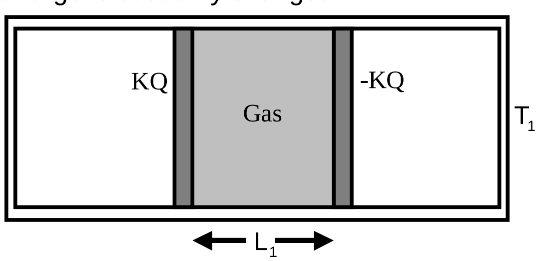
Sol. Stage 1



$P_1AL_1 = nRT_1 \qquad \frac{Q^2}{2A\varepsilon_0} = P_1A = \frac{nRT_1}{L_1}$ (Force Balancing)

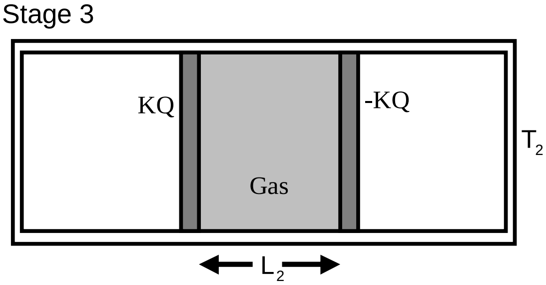
Stage 2

Charge is suddenly changed



System is not in equilibrium

Stage 3



L_2CL_1

Finally system in equilibrium

$P_2AL_2 = nRT_2 \qquad \frac{K^2Q^2}{2A\varepsilon_0} = P_2A = \frac{nRT_2}{L_2}$

Energy of the system is conserved between stage 2 and 3.

$\Rightarrow \frac{1}{2}\varepsilon_0E_2^2AL_1 + \frac{f}{2}nRT_1 = \frac{1}{2}\varepsilon_0E_2^2AL_2 + \frac{f}{2}nRT_2$

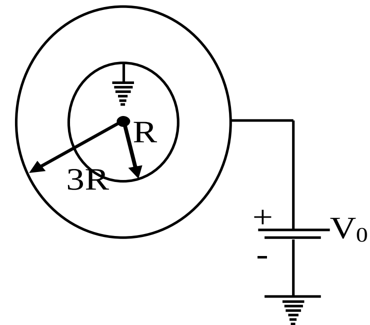
On solving all the above equation

$\frac{T_2}{T_1} = \frac{2K^2+f}{2+f} \qquad w_{ext} = \left(K^2-1\right)\frac{Q^2L_1}{2A\varepsilon_0} = \left(K^2-1\right)nRT_1$

Section – B (Maximum Marks: 12)

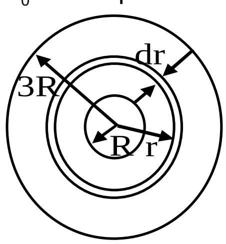
*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. A conducting sphere with inner radius R and outer radius 3R has some alloys mixed in it due to which its resistivity changes with radial distance r according to the function $\rho_r = Kr$, where K is a +ve constant. Inner surface of the sphere is grounded while outer surface of sphere in maintained at potential V_0 with the help of a battery of emf V_0. Battery is ideal. The potential of the sphere at radial distance of 2R is $V_0 \frac{\ell n\ a}{\ell n\ b}$. Find (a + b).



Ans. 5

Sol. $V_0 = I\ Req$



$$R_{eq} = \int dR$$

$$R_{eq} = \int \frac{\rho_r dr}{4\pi r^2}$$

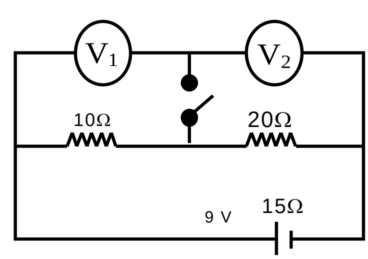
$$R_{eq} = \int_R^{3R} \frac{Kdr}{4\pi r} \Rightarrow R_{eq} = \frac{K}{4\pi} \ell n(3)$$

$$R(r) = \frac{K}{4\pi} \ln\left(\frac{r}{R}\right)$$

$$V(r) = IR(r) = \frac{V_0}{R_{eq}} \times R(r)$$

$$V(2R) = V_0 \frac{\ell n2}{\ell n3}$$

12. In the circuit shown in the figure the electromotive force of the battery is 9 V and its internal resistance is $15\,\Omega$. The two identical voltmeters can be considered ideal. Let V_1 and V_1' reading of 1st voltmeter when switch is open and closed respectively. Similarly, V_2 and V_2' be the reading of 2nd voltmeter when switch is open and closed respectively. Then find the value of $\frac{V_2' - V_2}{V_1 - V_1'}$.



Ans. 1

Sol. When switch is open

Current in circuit

$I = \frac{9}{45} A = 0.2A$

$V_1 + V_2 = 6V$ and because in series same current passes and voltmeters are identical so

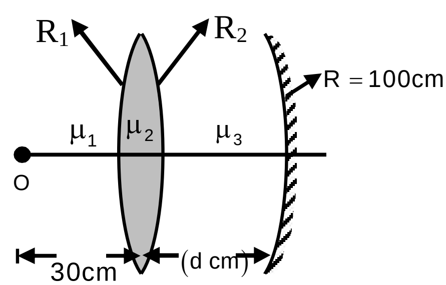
$V_1 = V_2 = 3V$

When switch closed

$V_1^{'} = 2V; V_2^{'} = 4V$

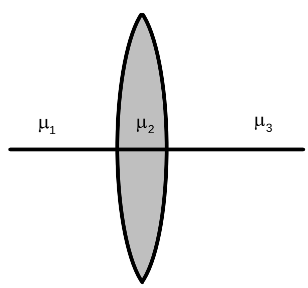
$\frac{V_2^{'} - V_2}{V_1 - V_1^{'}} = \frac{4-3}{3-2} = 1$

13. The given thin lenses have $R_1 = R_2 = 10cm$ and $\mu_1 = 1, \mu_2 = 2$ and $\mu_3 = 4$. If final image formed at object and value of 'd' is $x \times 10$ cm. Find 'x'.



Ans. 7

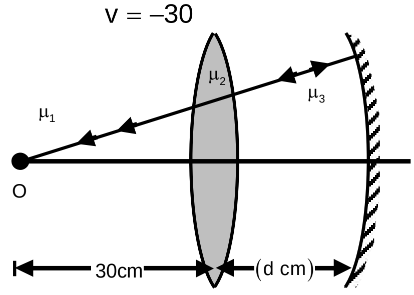
Sol.



$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_3 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$$

$$\frac{4}{v} - \frac{1}{-30} = \frac{2-1}{10} + \frac{4-2}{-10}$$

$$v = -30$$



$d + 30 = 100$

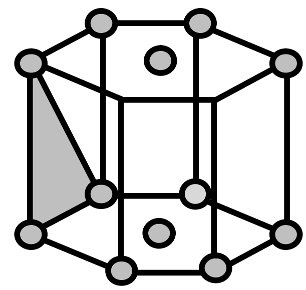
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** questions stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

There is a hollow prism with hexagonal base of side L_0 and height H_0 as shown. There are identical non conducting uniformly charged solid spheres are placed at each corner and two identical spheres at the centre of flat hexagonal faces. The charges spheres have uniform density of d_0 and radius of R_0. What will be the flux passing through the shaded region and average charge density (ϕ) hexagonal prism. It is given that the flux passing through the hexagonal flat surface is k times the total flux passing through the prism.



Take $\frac{R_0{}^3 d_0}{\varepsilon_0} = 1$ SI units

& $\frac{d_o R_0{}^3}{L_0^2 H_0} = 1$ SI units

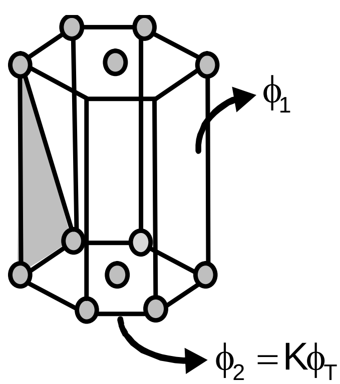
14. The value of ϕ_s is _________

Ans. 00000.35

15. The value of d is _________

Ans. 00004.83

Sol.



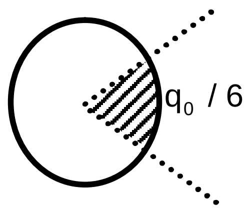
$\phi_s = \frac{\phi_1}{2}$

$6\phi_1 + 2\phi_2 = \phi_{total}$

$6\phi_1 + 2K\phi_{total} = \phi_{total}$

$\phi_1 = \frac{(1-2K)\phi_{total}}{6} \Rightarrow \phi_{total} = \frac{q_{in}}{\varepsilon_0}$

Top view



$$q_{in} = 12.\frac{q_0}{6} + 2\frac{q_0}{2}$$

$$\Rightarrow 3q_0 = q_{in} \Rightarrow 3d_0\frac{4}{3}\pi R_0^{\,3}$$

$$\phi_1 = \frac{(1-2k)}{6}\times\frac{4\pi R_0^{\,3}d_0}{\varepsilon_0}$$

$$\phi_s = \frac{(1-2k)}{3}\frac{\pi R_0^{\,3}d_0}{\varepsilon_0} = 0.35$$

Average charge density $= \frac{q_{in}}{V_{prism}}$

$$d = \frac{8\pi}{3\sqrt{3}}\frac{d_0R_0^{\,3}}{L_0^{\,2}W_0} = 4.83$$

Question Stem for Question Nos. 16 and 17

Question Stem

One side of a thin metal plate is illuminated by the sun. When the air temperature is T_0, the temperature of the illuminated side is T_1 and that of the opposite side is T_2. The temperature of each sides be $T_1^{'}$ & $T_2^{'}$, if another plate of double thickness is used. $\left(\text{Take } T_0 = 300\text{ K, } T_1\text{=330 K, } T_2 = 320\text{ K}\right)$

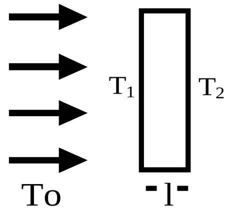
16. The value of $T_1^{'}$ (K) is___________

Ans. 00333.33

17. The value of $T_2^{'}$ (K) is___________

Ans. 00316.67

Sol. Left face
Power from sun = P



$$\frac{d\theta}{dt} \propto (T_1 - T_0) \qquad \frac{d\theta}{dt} = \lambda(T_1 - T_0)$$

$$\lambda = 4\sigma eAT_0^3$$

$P = \lambda(T_1 - T_0) + \frac{kA(T_1 - T_2)}{\ell}$ ....(i) (For left surface)

$\frac{KA(T_1 - T_2)}{\ell} = \lambda(T_2 - T_0)$ (For right surface)

When thickness is doubled

$P = \lambda(T_1' - T_0) + \frac{kA(T_1' - T_2')}{2\ell}$ .....(iii)

$\frac{KA(T_1' - T_2')}{2\ell} = \lambda(T_1' - T_0)$ ......(iv)

on solving

$$T_1' = \frac{2T_1^2 + T_1T_2 - T_0(3T_1 - T_2) - T_2^2}{2(T_1 - T_0)} = 333.33k.$$

$$T_2' = \frac{T_1T_2 + T_0(T_1 - 3T_2) + T_2^2}{2(T_1 - T_0)} = 316.67k.$$

Question Stem for Question Nos. 18 and 19

Question Stem

A boat is travelling in a river with a speed of 10 m/s along the stream flowing with a speed of 2 m/s. From this boat, a sound transmitter is lowered into the river through a rigid support. The wavelength of the sound emitted from the transmitter inside the water is 14.45 mm. Assume that attenuation of sound in water and air is negligible. The frequency detected by a receiver kept inside the river downstream is f_0 and the transmitter and the receiver are now pulled up into air. The air is blowing with a speed 5 m/sec in the direction opposite the river stream the frequency of the sound detected by the receiver is f_1.
(Temperature of the air and water = 20°C; Density of river water = 10^3 kg/m^3; Bulk modulus of the water = 2.088×10^9 Pa; Gas constant R = $8.31 J/mol-K$; Mean molecular mass of air = $28.8 \times 10^{-3} kg/mol$; Cp/Cv for air = 1.4)
Note: Boat velocity is with respect to ground & receiver is stationary w.r.t ground

18. The value of f_0 (KHZ) is__________

Ans. 00100.69

19. The value of f_1 (KHZ) is__________

Ans. 00103.04

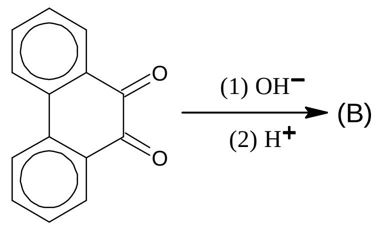
Sol. Use doppler effect.

Chemistry

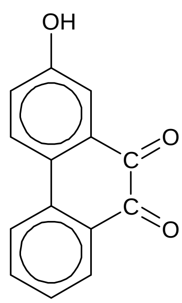
PART – II

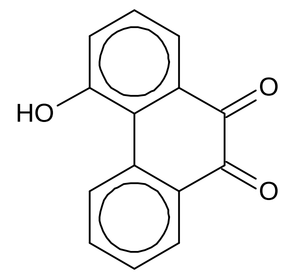
Section – A (Maximum Marks: 12)

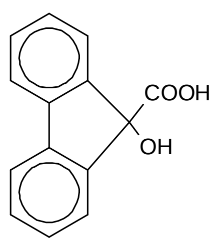
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

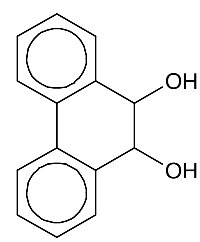
20. 

The compound (B) would be

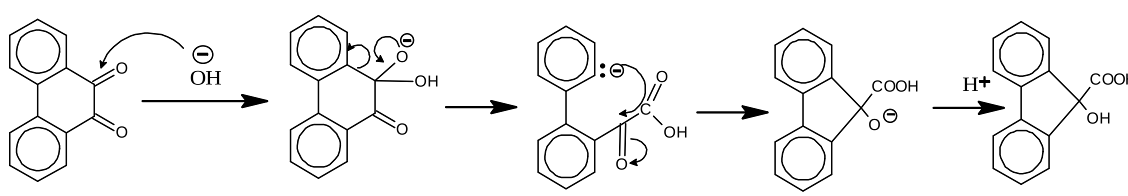
(A) 

(B) 

(C) 

(D) 

Ans. C

Sol. 

21. How many methoxy group present in the molecule of a compound (A) would be present by the result of a Zeisel method. Treatment of compound (A) $C_{20}H_{11}O_4N$ with hot conc. HI gives CH_3I indicating presence of methoxy group. When 4.24 mg of a compound (A) is treated with hot conc. HI and CH_3I thus produced is digested with alc. $AgNO_3$, 11.62 mg of AgI is obtained.

(A) 1
(B) 2
(C) 3
(D) 4

And. D

Sol. $C_{20}H_{11}O_4N + HI \longrightarrow CH_3I \xrightarrow{AgNO_3} AgI$

$\frac{1.24}{329} = .013$ $\frac{11.62}{235} = 0.05$

No. of mole of $AgI \approx 4\times$ mole of $C_{20}H_{11}ON$

22. The blue colour produced on adding H_2O_2 to acidified $K_2Cr_2O_7$ solution and shaking with ether is due to the formation of

(A) Cr_2O_3

(B) CrO_4^{-2}

(C) CrO_3

(D) CrO_5

Ans. D

Sol. $K_2Cr_2O_7 + H_2O_2 + H_2SO_4 \xrightarrow{ether} \underset{(Blue)}{CrO_5}$

23. A 0.001 molal solution of a complex [MA_8] in water has the freezing point -0.0054^oC. Assuming 100% ionization of the complex salt and K_f for H_2O = 1.86 K-Kgmol^{-1}, write the correct representation of the complex.

(A) $[MA_8]$

(B) $[MA_7]A$

(C) $[MA_6]A_2$

(D) $[MA_5]A_3$

Ans. C

Sol. $\Delta T_f = i\,K_f \frac{w\times 1000}{M.W\times W} = i\,K_f m$

$.0054 = i\ 1.86\times.001$

$i = \frac{.0054}{1.86\times.001} \approx 2.90$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

24. A cation which form am amine complex ion with excess of NH_3 is

(A) Ag^{+}

(B) Cu^{+2}

(C) Co^{+2}

(D) Al^{+3}

Ans. ABC

Sol. $Ag^{+} + 2NH_3 \longrightarrow \left[Ag\left(NH_3\right)_2\right]^{+}$

$Cu^{+2} + 4NH_3 \longrightarrow \left[Cu\left(NH_3\right)_4\right]^{+2}$

$Co^{+2} + 6NH_3 \longrightarrow \left[CO\left(NH_3\right)_6\right]^{+2}$

25. Which of the following statements regarding P_4O_{10} is correct?

(A) It is an anhydride of metaphosphoric acid

(B) It is a strong dehydrating agent

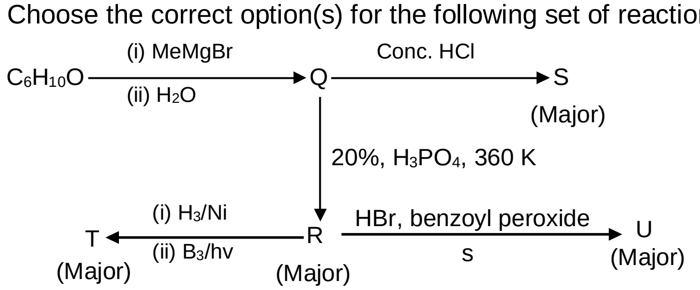
(C) In P_2O_5, each phosphorous is attached to three oxygen atoms through normal covalent bond and also with another oxygen atom through a coordinate bond.

(D) The dehydrated product of HNO_3 obtained by treating it with P_4O_{10} is NO_2.

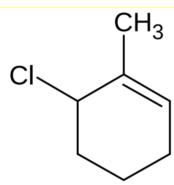
Ans. ABC

Sol. HNO_3 on dehydration gives N_2O_5

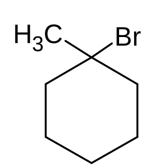
26. Choose the correct option(s) for the following set of reactions



(A)

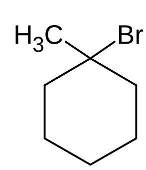


(S)

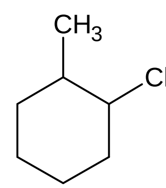


(T)

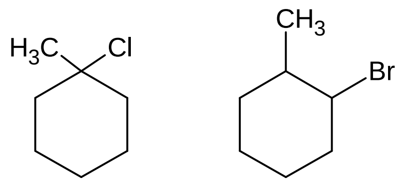
(B)



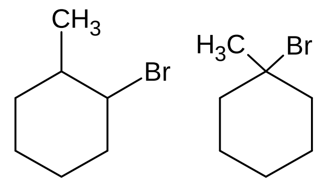
(U)



(S)

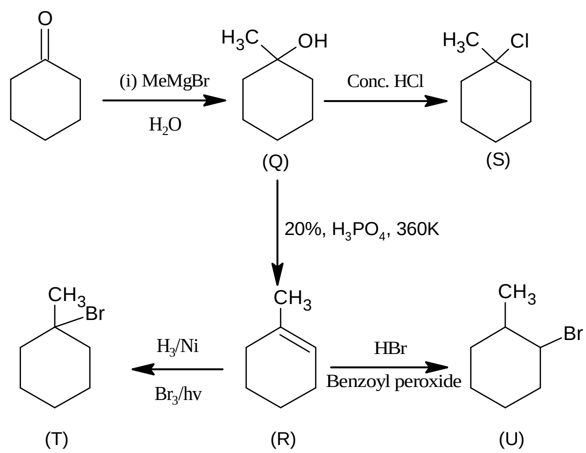
(C) 

(S) (U)

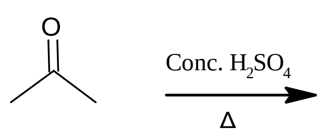
(D) 

(U) (T)

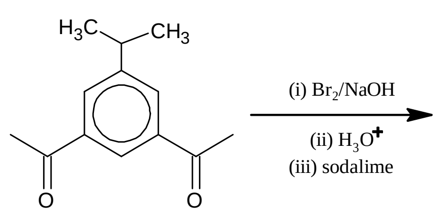
Ans. C

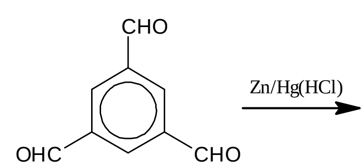
Sol. 

27. The reaction(s) leading to the formation of 1, 3, 5 trimethyl benzene is/are

(A) 

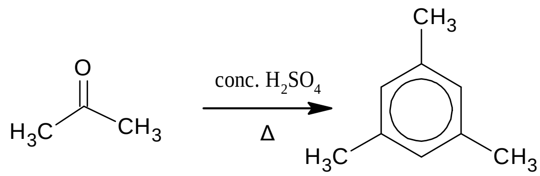
(B) $Me-C\equiv CH \xrightarrow[373\ K]{\text{heated iron tube}}$

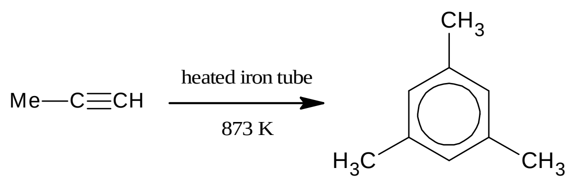
(C) 

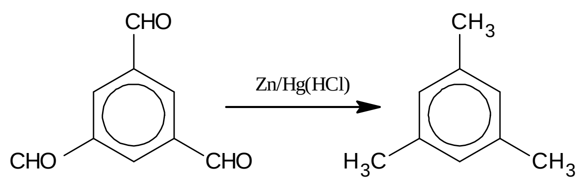
(D) 

Ans. ABD

Sol.







28. Which of the following statement(s) is/are correct?

(A) If temperature coefficient is greater than zero, cell reaction is endothermic.

(B) If temperature coefficient is less than zero, cell reaction is endothermic.

(C) If temperature coefficient is less than zero, cell reaction is exothermic.

(D) If E_{cell} is negative then ΔG is negative and cell reaction is spontaneous.

Ans. AC

Sol. $\left(\frac{dE}{dT}\right)=\frac{E_2-E_1}{T_2-T_1}$, According to Nernst equation, on $\uparrow$ in temp. E.M.F $\downarrow$ es.

29. In which of the following pair of solution will the values of Vant Hoff factor be same? (Assuming same dissociation)

(A) $0.05\text{ M } K_4\left[Fe(CN)_6\right]$ and $0.10\text{M } FeSO_4$

(B) $0.10\text{ M } K_4\left[Fe(CN)_6\right]$ and $0.05\text{ M } FeSO_4.(NH_4)_2SO_4.6H_2O$

(C) 0.20 M NaCl and 0.10 $BaCl_2$

(D) $0.05\text{ M } FeSO_4.(NH_4)_2SO_4.6H_2O$ and $0.02\text{M } KCl.MgCl_2.6H_2O$

Ans. BD

Sol. $K_4\left[Fe(CN)_6\right]\longrightarrow 4K^+ + \left[Fe(CN)_6\right]^{-4}$

$i=5$ No. of ions

$FeSO_4.(NH_4)_2SO_4.6H_2O\longrightarrow Fe^{+2}+2NH_4^+ + 2SO_4^{-2}$

i = 5 = No. of ions

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. Find the quantum numbers of the excited state of electrons (Let n_1 and n_2) in He^+ ion which on transition to ground state emit two photons of wavelength 30.4 nm and 108.5 nm respectively. Find the sum of $(n_1 + n_2) = ?$

Ans. 7

Sol. $\frac{1}{\lambda} = R_H (z)^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

for $n_1 = 1 \quad n_2 = ?$

$$\frac{1}{30.4 \times 10^{-9}} = 1.09678 \times 10^7 \times 4\left[\frac{1}{1^2} - \frac{1}{n_2^2}\right]$$

$n_2 = 2$

for $n_1 = 2$ (first excited state), $n_2 = ?$

$$\frac{1}{108.5 \times 10^{-9}} = 1.09678 \times 10^7 \times 4\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$$

$\boxed{n_2 = 5}$

sum of n_1 & $n_2 = 2 + 5 = 7$

31. A compound AB forms NaCl type crystals and another compound XY forms CsCl type crystals. The formula. Mass of XY is three times that of AB but the cubic edge length of unit cell of AB crystals is twice that of XY. Calculate the ratio of density of XY to that of AB.

Ans. 6

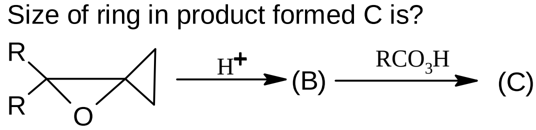
Sol. Density of AB = $\frac{4 \times M_1}{N_{a_r} \times \ell_1^3}$ ...(i)

Let molar mass of AB be M_1 gm/mol and edge length of unit cell be ℓ_1(m).

Density of XY $= \frac{1 \times 3M_1}{N_{a_r} \times \left(\frac{\ell_1}{2}\right)^3} = \frac{8 \times 3 \times M_1}{N_{a_r} \times \ell_1^3}$ ...(ii)

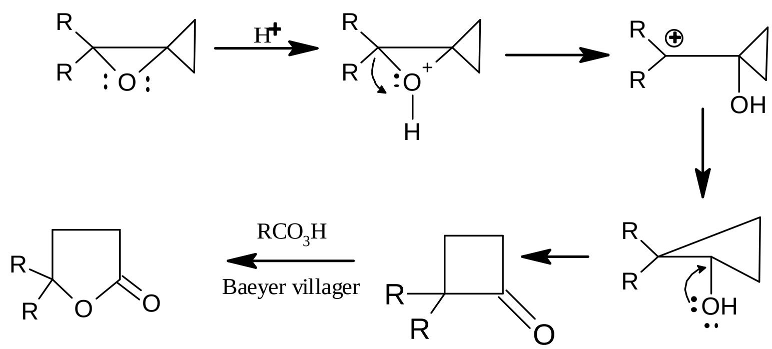
(ii) /(i) = 6

32. Size of ring in product formed C is?



Ans. 5

Sol.



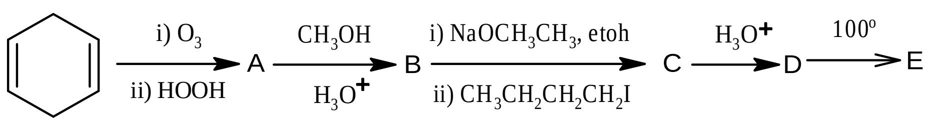
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** questions stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

For the following reaction scheme



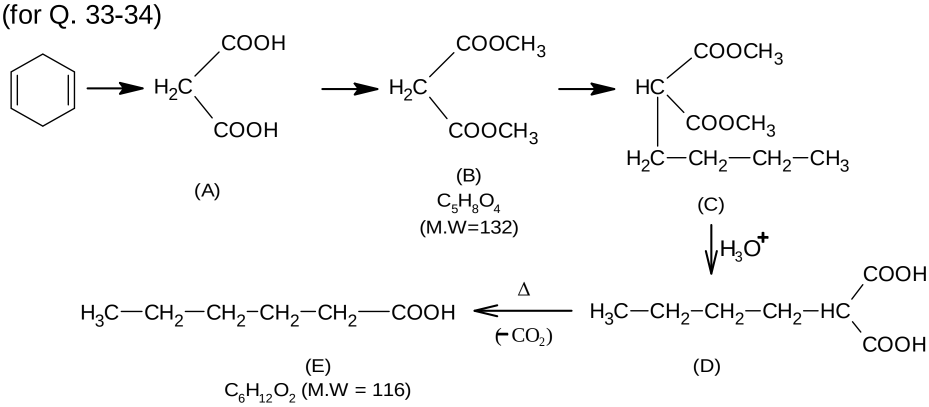
33. The molecular wight of B is ……………

Ans. 00132.00

34. The molecular weight of E is…………

Ans. 00116.00

Sol. (for Q. 33-34)



Question Stem for Question Nos. 35 and 36

Question Stem

A 10^{-3} molal aqueous solution of a chromium complex having same number of ammonia molecule and Cl ions freezes at $-0.00558^{\circ}C$. If all the ammonia molecules are acting as ligands and the percent of hydrogen in the complex is 4.6 (Given $K_f(H_2O) = 1.86 K\ Kgmol^{-1}$) and assume 100% ionization self.

35. Find the number of Cl atom is complex is___.

Ans. 00004.00

36. The value of Vant Hoff factor is______.

Ans. 00003.00

Sol. (for Q. 35-36)

Let the molecular formula of the complex is $Cr(NH_3)_x Cl_x$

$\therefore$ % H is in the complex

$$\frac{3x}{52+17x+35.5x} \times 100 = 4.6$$

$x \approx 4.08 \approx 4$

Now, $\Delta T_f = i\ k_f m$

$$i = \frac{.00558}{1.86 \times 10^{-3}} \approx 3$$

Question Stem for Question Nos. 37 and 38

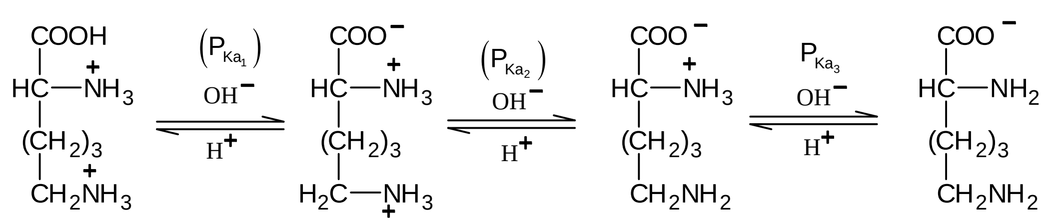
Question Stem

The isoelectric point (P_I) is the P_H at which the amino acid exists only as a dipolar ion with net charge zero. Structure of lysine and aspartic acid are $CO\overset{\ominus}{O}CH\left(\overset{+}{N}H_3\right)(CH_2)_4NH_2$ and $CO\overset{\ominus}{O}CH\left(\overset{+}{N}H_3\right)CH_2COOH$ respectively. The P_{Ka_1}, P_{Ka_2} and P_{Ka_3} of the dictation lysine are 2.18, 8.95 and 10.53 respectively. The P_{Ka_1}, P_{Ka_2} and P_{Ka_3} of the cation of aspartic acid are 1.88, 3.65 and 9.60 respectively.

37. Isoelectric point lysine is__________.

Ans. 00009.74

Sol. The dissociation of cationic form of lysine can be represented.



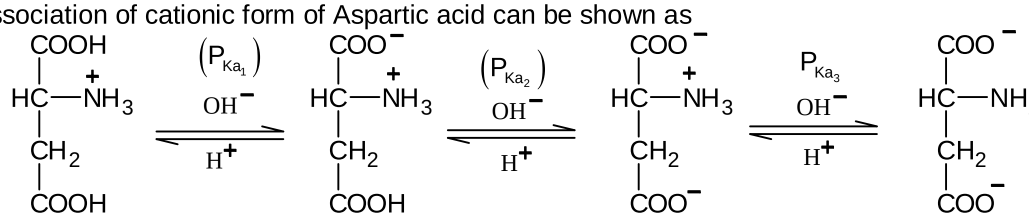
Net change (+2) (+1) (0) (–1)

$$P_I = \frac{P_{Ka_2} + P_{Ka_3}}{2} = \frac{8.95 + 10.53}{2} = 9.74$$

38. Isoelectric point of Aspartic acid is __________.

Ans. 00002.77

Sol. The dissociation of cationic form of Aspartic acid can be shown as



Net change (+1) (0) (–1) (–2)

$$P_I = \frac{P_{Ka_2} + P_{Ka_2}}{2} = \frac{1.88 + 3.65}{2} = 2.77$$

Mathematics PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

39. Polynomial function f (x) satisfying the condition $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$. If f (10) = 1001, then f(20) is

(A) 7001

(B) 8001

(C) 8000

(D) None of these

Ans. B

Sol. $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$

let $f(x)=a_nx^n+a_{n-1}x^{n-1}+a_{n-2}x^{n-2}+.....+a_ax+a_0$

$$f\left(\frac{1}{x}\right)=\frac{a_n}{x^n}+\frac{a_{n-1}}{x^{n-1}}+a_{n-2}\cdot\frac{1}{x^{n-2}}+..........+\frac{n}{x}+a_0$$

$$f(x)\cdot f\left(\frac{1}{x}\right)=a_0a_nx^n+\left(a_{n-1}a_0+a_na_1\right)x^{n-1}+x^{n-2}\left(a_0a_{n-2}+a_1a_{n-1}+a_2a_n\right)$$

$$f(x)+f\left(\frac{1}{x}\right)=a_nx^n+a_{n-1}x^{n-1}+x^{n-2}a_{n-2}+........+a_1x+a_0$$

$$+\frac{a_n}{x^n}+\frac{a_{n-1}}{x^{n-1}}+\frac{a_{n-2}}{x^{n-2}}+.......+\frac{a_1}{x}+a_0$$

$a_n=1\quad a_0=1;\ \{a_1=a_2=a_3=......=a_{n-1}=0\}$

$f(x)=x^n+1$

$f(10)=10^n+1=10^3+1$

n = 3

$f(x)=x^3+1$

$f(20)=(20)^3+1=8001$

40. Let $f(x)=e^{\left\{e^{|x|}\operatorname{sgn}x\right\}}$ and $g(x)=e^{\left[e^{|x|}\operatorname{sgn}x\right]}$, $x\in R$ where $\{\ \}$ and $[\ ]$ denotes the fractional and integral part functions, respectively. Also $h(x)=\log(f(x))+\log(g(x))$ then for real x, $h(x)$ is

(A) An odd function

(B) An even function

(C) Neither odd nor an even function

(D) Both odd as well as even function

Ans. A

Sol. $h(x) = \log(f(x).\, g(x)) = \log\, e^{\{y\}+[y]} = \{y\} + [y] = e^{|x|}\operatorname{sgn} x$

$$\therefore h(x) = e^{|x|}\operatorname{sgn}\, x = \begin{cases} e^{x}, & x > 0 \\ 0, & x = 0 \\ -e^{x}, & x < 0 \end{cases}$$

$$\Rightarrow h(-x) = \begin{cases} e^{-x}, x < 0 \\ 0,\ x = 0 \\ -e^{x}\ x > 0 \end{cases} \Rightarrow h(x) + h(-x) = 0 \text{ for all x.}$$

41. The length and foot of the perpendicular from the point (2, -1, 5) to the line $\frac{x-11}{10} = \frac{y+2}{-4} = \frac{z+8}{-11}$ are

(A) $\sqrt{14}, (1, 2, -3)$

(B) $\sqrt{14}, (1, -2, 3)$

(C) $\sqrt{14}, (1, 2, 3)$

(D) None of these

Ans. C

Sol. Let co-ordinates of foot of perpendicular are (a, b, c)
Since $\frac{a-11}{10} = \frac{b+2}{-4} = \frac{c+8}{-11} = k$
Then $a = 10k + 11, b = -4k - 2, c = -11k - 8$
Here all alternative give a = 1, then k = -1. So foot is (1,2,3) and perpendicular distance between (1,2,3) and (2,-1,5) is $\sqrt{14}$.

42. If x = 9 is the chord of contact of the hyperbola $x^2 - y^2 = 9$, then the equation of the corresponding pair of tangent is

(A) $9x^2 - 8y^2 + 18x - 9 = 0$

(B) $9x^2 - 8y^2 - 18x + 9 = 0$

(C) $9x^2 - 8y^2 - 18x - 9 = 0$

(D) $9x^2 - 8y^2 + 18x + 9 = 0$

Ans. B

Sol. Equation of chord of contact
$xh - yk - 9 = 0$ ....(1)
$x - 9 = 0$ .....(2)
$h = 1$ and $k = 0$
pair of tangent
$SS' = T^2$
$\left(x^2 - y^2 - 9\right)\left(-8\right) = \left(x - 9\right)^2 \quad 9x^2 - 8y^2 - 18x + 9 = 0$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

43. Which of the following options are correct?

(A) $\left[({}^nC_0 + {}^nC_3 + {}^nC_6 + ...) - \frac{1}{2}({}^nC_1 + {}^nC_2 + {}^nC_4 + {}^nC_5 + ...)\right]^2 + \frac{3}{4}\left({}^nC_1 - {}^nC_2 + {}^nC_4 - {}^nC_5 + ...\right)^2 = 1$

(B) If a and b are two positive numbers such that $a^5b^2 = 4$ then the maximum value of $\log_{2^{1/5}}\left(a^2\right)\cdot\log_{2^{1/2}}\left(b^2\right)$ is equal to 4

(C) Constant team in $((((((x-2)^2 - 2)^2 - 2)^2 - 2)^2 - 2)^2 ....2)^2$ is equal to 2

(D) The coefficient of x^{24} in $\left(\frac{{}^{25}C_1}{{}^{25}C_0} - x\right)\left(x - 2^2\frac{{}^{25}C_2}{{}^{25}C_1}\right)\left(x - 3^2\frac{{}^{25}C_3}{{}^{25}C_2}\right)\left(x - 4^2\frac{{}^{25}C_4}{{}^{25}C_3}\right).....$ $\left(x - 25^2\frac{{}^{25}C_{25}}{{}^{25}C_{24}}\right)$ is equal to 2925

Ans. ABD

Sol. (A) $\left(1+x\right)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + ... + {}^nC_nx^n$

Put $x = \omega$ where $\omega = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$

$\left({}^nC_0 + {}^nC_3 + {}^nC_6 + ...\right) + ... + \left({}^nC_1 + {}^nC_4 + {}^nC_7 + ...\right)\omega + \left({}^nC_2 + {}^nC_5 + {}^nC_8 + ...\right)\omega^2 = \left(1+\omega\right)^n$

$\left({}^nC_0 + {}^nC_3 + {}^nC_6 + ...\right) + \left({}^nC_1 + {}^nC_4 + {}^nC_7 + ...\right)\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) + \left({}^nC_2 + {}^nC_5 + {}^nC_8 + ...\right)\left(-\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)$

$\left({}^nC_0 + {}^nC_3 + {}^nC_6 + ...\right) - \frac{1}{2}\left({}^nC_1 + {}^nC_2 + {}^nC_4 + {}^nC_5 + ...\right) + \frac{i\sqrt{3}}{2}\left({}^nC_1 - {}^nC_2 + {}^nC_4 - {}^nC_5 + ...\right) = \left(-\omega^2\right)^n$

(B) $a^5b^2 = 4 \Rightarrow 5\log_2\left(a\right) + 2\log_2\left(b\right) = 2$

$y = 10\log_2 a \cdot 4\log_2 b = 40\log_2 a \cdot \log_2 b$

$\text{A.M.} \ge \text{G.M.} \Rightarrow \frac{5\log_2\left(a\right) + 2\log_2\left(b\right)}{2} \ge \sqrt{10\log_2 a \cdot \log_2 b}$

$\log_2 a \cdot \log_2 b \le \frac{1}{10} \Rightarrow y \le 4, y = 4$ when $5\log_2 a = 2\log_2 b \Rightarrow a^5 = b^2 = 2$

(C) For constant term put x = 0 and get constant term = 4
(D) Coefficient of $x^{24} = 1.25 + 2.24 + 3.23 + ... + 25.1 = 2925$

44. $a_1, a_2, a_3, ...$ are distinct terms of an A.P. We call (p,q,r) an increasing triad if a_p, a_q, a_r are in G.P. where $p, q, r \in N$ such that $p < q < r$. If (5, 9, 16) is an increasing triad, then which of the following options is/are correct

(A) If a_1 is a multiple of 4 then every term of the A.P. is an integer

(B) $(85, 149, 261)$ is an increasing triad

(C) If the common difference of the A.P. is $\frac{1}{4}$, then its first term is $\frac{1}{3}$

(D) ratio of (4k + 1)th term and 4kth term can be 4

Ans. ABC

Sol. Let R be the common ratio of the G.P. and D be the common difference of A.P.

$a_5 = a_5, a_9 = Ra_5, a_{16} = a_5R^2$

$a_9 - a_5 = 4D \Rightarrow (R-1)a_5 = 4D$

$a_{16} - a_9 = 7D \Rightarrow R(R-1)a_5 = 7D$

From equation (1)/(2), we get $\frac{1}{R} = \frac{4}{7} \Rightarrow R = \frac{7}{4}$

From equation (2) – (1), we get $(R-1)^2 a_5 = 3D \Rightarrow \frac{9a_5}{16} = 3D$

$\frac{3}{16}(a_1 + 4D) = D \Rightarrow \frac{3}{16}a_1 = \left(1 - \frac{3}{4}\right)D \Rightarrow a_1 = \frac{4D}{3}$

45. If $\int \frac{e^{x-1}}{(x^2 - 5x + 4)} 2x\,dx = AF(x-1) + BF(x-4) + C$ and $F(x) = \int \frac{e^x}{x} dx$, then

(A) $A = -2/3$

(B) $B = (4/3)e^3$

(C) $A = 2/3$

(D) $B = (8/3)e^3$

Ans. AD

Sol. $\frac{2x}{(x-1)(x-4)} = \frac{C}{x-1} + \frac{D}{x-4}$

$2x = C(x-4) + D(x-1)$

$\therefore \quad C = -2/3,\ D = 8/3$

$$\therefore \quad \int \frac{e^{x-1}}{(x-1)(x-4)} 2x\, dx = \int e^{x-1}\left(\frac{-2/3}{x-1} + \frac{8/3}{x-4}\right) dx$$

$$= -\frac{2}{3}F(x-1) + \frac{8}{3}e^3F(x-4) + C$$

$$\therefore \quad A = -2/3,\ B = 8/3\ e^3$$

46. The vertices of a triangle ABC are $A \equiv (2,0,2)$, $B(-1,1,1)$ and $C \equiv (1,-2,4)$. The points D and E divide the sides AB and CA in the ratio 1:2 respectively. Another point F is taken in space such that the perpendicular drawn from F to the plane containing ΔABC, meets the plan at the point of intersection of the line segment CD and BE. If the distance of F from the plane of triangle ABC is $\sqrt{2}$ units, then

(A) the volume of the tetrahedron ABCF is $\frac{7}{3}$ cubic units

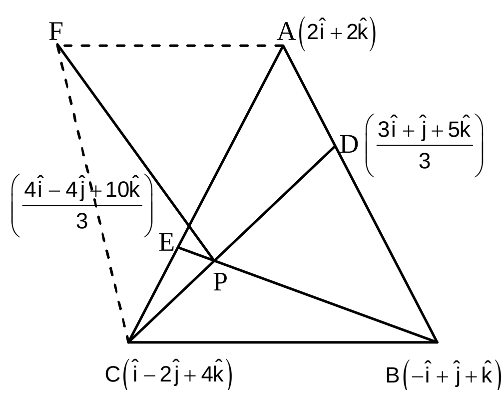
(B) the volume of the tetrahedron ABCF is $\frac{7}{6}$ cubic units

(C) one of the equation of the line AF is $\vec{r}\left(2\hat{i} + 2\hat{k}\right) + \lambda\left(2\hat{k} - \hat{i}\right)(\lambda \in R)$

(D) one of the equation of the line AF is $\vec{r} = \left(2\hat{i} + 2\hat{k}\right) + \mu\left(\hat{i} + 7\hat{k}\right)$

Ans. AC

Sol.



$$CD : \vec{r} = \left(\hat{i} - 2\hat{j} + 4\hat{k}\right) + \frac{\lambda}{3}\left(7\hat{j} - 7\hat{k}\right)$$

$$BE : \vec{r} = \left(-\hat{i} + \hat{j} + \hat{k}\right) + \frac{\mu}{3}\left(7\hat{i} - 7\hat{j} + 7\hat{k}\right)$$

$P \equiv (\hat{i} - \hat{j} + 3\hat{k})$

Area of tetrahedron ABCF

$$= \frac{1}{3}(\text{Area of base triangle}) \times \text{height} = \frac{7}{3}\text{cubic unit}$$

$$\vec{AB} \times \vec{AC} = 7\hat{j} + 7\hat{k}, \left|\vec{PF}\right| = PF = \sqrt{2} \text{ units}$$

$$\vec{PF} = \sqrt{2}\left(\frac{7\hat{j}+7\hat{k}}{\sqrt{49+49}}\right) = \hat{j}+\hat{k} = \text{P.V. of F-P.V. of P}$$

P.V. of F=$\hat{i}$+4$\hat{k}$

Vectors equation of AF is $\vec{r}\,2\left(\hat{i}+\hat{j}\right)+\alpha\left(-\hat{i}+2\hat{k}\right)$

47. Let the equation of a straight line L in complex form be $a\bar{z}+\bar{a}z+b=0,$ where a is a complex number and b is a real number, then

(A) the straight line $\frac{z-c}{a}+\frac{i\left(\bar{z}-\bar{c}\right)}{a}=0$ makes an angle of 45° with L and passes through a point c (where c is a complex number)

(B) the straight line $\frac{z-c}{a}=\frac{i\left(\bar{z}-\bar{c}\right)}{a}$ makes an angle of 45° with L and passes through a point c (where c is a complex number)

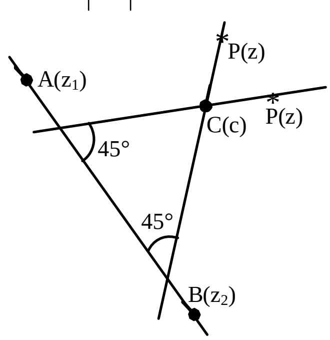
(C) the complex slope of the line L is $-\frac{a}{a}$

(D) the complex slope of the line L is $\frac{a}{a}$

Ans. ABC

Sol. Let P(z) be any point on the required line.

Then, $\frac{\vec{CP}}{\left|\vec{CP}\right|}$ i.e. $\frac{z-c}{|z-c|}$ is a unit vector parallel to it



Let A(z_1) and B(z_2) be two points on

$\bar{a}z+a\bar{z}+b=0$ then $\frac{z_2-z_1}{|z_2-z_1|}$ is a unit vector parallel to the line

$$a\bar{z}+\bar{a}z+b=0$$

$$\frac{z-c}{|z-c|}=\frac{z_2-z_1}{|z_2-z_1|}e^{\pm\left(\frac{\pi}{4}\right)}$$

$$\frac{(z-c)^2}{(z-c)(\bar{z}-\bar{c})}=\frac{(z_2-z_1)^2}{(z_2-z_1)(\bar{z}_2-\bar{z}_1)}e^{\pm i\frac{\pi}{2}}$$

$\frac{z-c}{z-c} = \pm i\left(\frac{z_2 - z_1}{z_2 - \bar{z}_1}\right)$

$\because$ $A(z_1)$ and $B(z_2)$ are on the line $a\bar{z} + \bar{a}z + b = 0$ therefore $a\bar{z}_1 + \bar{a}z_1 + b = 0$

$a\bar{z}_2 + \bar{a}z + b = 0$

$\Rightarrow -\frac{a}{a} = \frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}$

From equation (1) and (2) we get $\frac{z-c}{a} \pm \frac{i(\bar{z} - \bar{c})}{\bar{a}} = 0$

48. The function $f(x) = x^{1/3}(x-1)$

(A) has two inflection points

(B) has one point of extremum

(C) is non – differentiable

(D) Range of $f(x)$ is $\left[-3 \times 2^{-8/3}, \infty\right)$

Ans. ABCD

Sol. $f(x) = x^{1/3}(x-1)$

$\Rightarrow \frac{df(x)}{dx} = \frac{4}{3}x^{1/3} - \frac{1}{3}.\frac{1}{x^{2/3}} = \frac{1}{3x^{2/3}}[4x-1]$

$\Rightarrow f(x)$ changes sign from –ve to +ve, $x = 1/4$, which is point of minima.

Also $f(x)$ does not exist at $x = 0$ as $f(x)$ has vertical tangent at $x = 0$.

$f''(x) = \frac{4}{9}.\frac{1}{x^{2/3}} = \frac{1}{3}.\frac{2}{3}.\frac{1}{x^{5/3}} = \frac{2}{9x^{2/3}}\left[2 + \frac{1}{x}\right] = \frac{2}{9x^{2/3}}\left[\frac{2x+1}{x}\right]$

$\therefore f''(x) = 0$ at $x = -\frac{1}{2}$ which is the point of inflection at $x = 0$, $f''(x)$ does not exists but $f''(x)$ changes sign, hence $x = 0$ is also the point of inflection.

From the above information the graph of $y = f(x)$ is as shown.

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. A, B, C and D cut a pack of 52 cards successively in the order given. If the person who cuts a spade first receives Rs. 350 and if the expectation of A is λ then $\left[\frac{\lambda}{64}\right]$ is equal to (where [.] denotes greatest integer function)_________

Ans. 2

Sol. E = even of any one cutting a space in one cut

$n(E) = {}^{13}C_1$
$n(S) = {}^{52}C_1$
$P(E) = 1/4 = p, P(\bar{E}) = q$
Probability of a winning $= p + qqqp + qqqqqqp + ...\infty$
$= \frac{P}{1-q^3} = \frac{64}{175} = 128$
$\left[\frac{\lambda}{64}\right] = 2$

50. Let $f(x) = \lim\limits_{n\to\infty} \frac{x^{2n-1} + ax^2 + bx}{x^{2n} + 1}$. If $f(x)$ is continuous for all $x \in R$, then the value of $a + 8b$ is

Ans. 8

Sol. $f(x)\begin{bmatrix} ax^2 + bx & \text{for } -1 < x < 1 \\ \frac{a-b-1}{2} & x = -1 \\ \frac{a+b+1}{2} & x = 1 \\ \frac{1}{x} & \text{for } x > 1 \text{ or } x < -1 \end{bmatrix}$

for continuity at $x = 1$ we have $a + b = \frac{a+b+1}{2}$
hence, $a + b = 1$ (1)
for continuity at $x = -1$
$a - b = -1$ $a - b = -1$ (2)
hence $a = 0$ and $b = 1$

51. A and B are two independent events. The probability that both A and B occur is 1/6 and the probability that at least one of them occurs is 2/3. Find $8P(A) + 9P(B)$ if $P(A) > P(B)$.

Ans. 7

Sol. $P(A \cap B) = P(A)P(B) \Rightarrow 1/6 = P(A)P(B)$
$P(A \cup B) = P(A) + P(B) \cdot P(A \cap B)$
$\Rightarrow 2/3 = P(A) + \frac{1}{6P(A)} - \frac{1}{6}$
$\Rightarrow 6(P(A))^2 - 5P(A) + 1 = 0 \Rightarrow P(A) = 1/2, P(B) = 1/3$
So, $8P(A) + 9P(B) = 4 + 3 = 7$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** questions stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Consider the locus of the complex number z in the Argand plane given by $\text{Re}(z)-2=|z-7+2i|$.

Let $P|z_1|.Q(z_2)$. Be two complex numbers satisfying the given locus and also satisfying

$\arg\left(\frac{z_1-(2+\alpha i)}{z_2-(2+\alpha i)}\right)=\frac{\pi}{2}(\alpha\in R)$.

52. The minimum value of PR.QR where R represents the point (7, -2) is

(A) 25

(B) 12

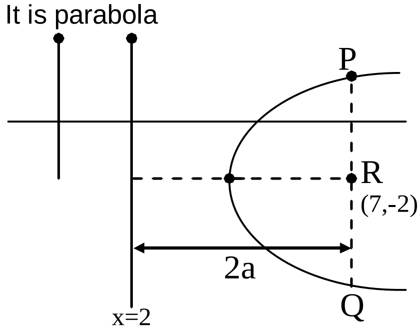
(C) 10

(D) $\sqrt{50}$

Ans. 00025.00

Sol. $(x-2)=\sqrt{(x-7)^2+(y+2)^2}$

It is parabola



$(PR)(RQ)=4a^2$

$=4\times\left(\frac{5}{2}\right)^2$

$=25$

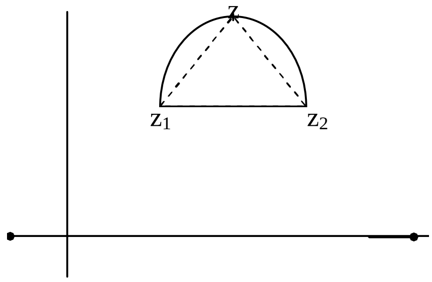
53. $\arg\left(\frac{z_1-(7-2i)}{z_1-(7-2i)}\right)$ equals

Ans. 00001.57

Sol. $\text{Re}\,z-2=|z-7+2i|$

Let $P(z_1), Q(z_2)$

$$\arg\left(\frac{z_1-(2+\propto i)}{z_2-(2+\propto i)}\right)=\frac{\pi}{2}(\propto \varepsilon R)$$



Question Stem for Question Nos. 54 and 55

Question Stem

T is the region of the plane x + y + z = 1 with x, y, z >0. S is the set of points (a, b, c) in T such that just two of the following three inequalities hold: $a \le \frac{1}{2}, b \le \frac{1}{3}, c \le \frac{1}{6}$.

54. Area of the region T is

Ans. 00000.86

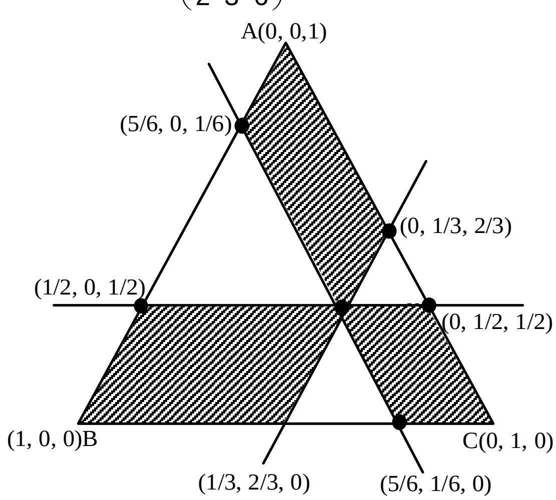
Sol. T is an equilateral triangle with the vertices at (1, 0, 0), (0, 1, 0) and (0, 0, 1)

$\Rightarrow$ area of the region T is $\frac{\sqrt{3}}{2}$.

55. Area of the region S is

Ans. 00000.34

Sol. Take a point $P\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$ on the plane x + y + z = 1



The region S is shown as shaded region.

Area of $S = \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{4}\left(a^2 + b^2 + c^2\right)$, where a, b, c are sides of the small equilateral triangles

$= \frac{7\sqrt{3}}{36}$.

Question Stem for Question Nos. 56 and 57

Question Stem

Consider the line $\frac{x-1}{2} = \frac{y}{-1} = \frac{z+1}{2}$ and the point $C(-1, 1, 2)$. Let D be the image of C in the line.

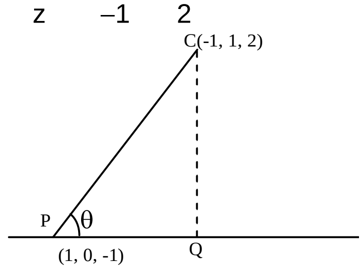
56. The distance C from the line is

Ans. 00003.73

57. The distance of the origin of the plane through the line and the point C is

Ans. 00000.45

Sol. $\frac{x-2}{z} = \frac{y}{-1} = \frac{z+1}{2}$



$CQ = CP\sin\theta$

$= \frac{\left|\vec{CP} \times \vec{L}\right|}{\left|\vec{L}\right|}$

$= \frac{\left\|\begin{matrix} i & \hat{j} & \hat{k} \\ 2 & -1 & -3 \\ 2 & -1 & 2 \end{matrix}\right\|}{\sqrt{4+1+4}} = \frac{\sqrt{125}}{3}$